

Optimise Your Test System Smartly

As higher targets are set to increase profits and decrease operating costs, test engineers are subjected to the ever-increasing stress of improving the efficiency of their testing operations and ultimately getting higher throughput for the same number of work hours. Read on to find out some simple ways to enhance your testing productivity



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Chips are becoming faster and more efficient keeping with Moore's Law. This is pushing consumer electronics technologies to new heights while making them all the more complex. Smarter test equipment are required to keep up with the new technology. Also, test engineers need to improvise techniques in order to enhance the productivity of their operations. Here is how you can improve your test performance.

Upgrade embedded controller, lower measurement time

As Intel and AMD come up with newer microarchitectures on their processors, there is a definite impact on the system controller and, in turn, the testing performance and productivity. The just released Ivy Bridge

microarchitecture—a 22nm die-shrink of the previous Sandy Bridge microarchitecture—brings in a plethora of new features that enhance computing performance.

If you go for an embedded con-

troller that features a processor with a large cache, the controller wouldn't have to access the comparatively slower DDR3 RAM for that data. The lower latency of the cache will, in turn, improve processor performance, delivering faster measurements and results. This is especially true for operations that require intensive signal and data processing.

"Especially for RF measurements and RF protocol testing, CPU performance is often the single most significant factor preventing faster measurement performance. While actual system performance can depend on a variety of factors such as memory available and other applications running in the background, a strong correlation exists between the CPU performance and measurement time for automated test systems," explains David Hall, product manager for RF and Communications at the National Instruments Dev Zone.

Fig. 1 shows how the PXIe-8106

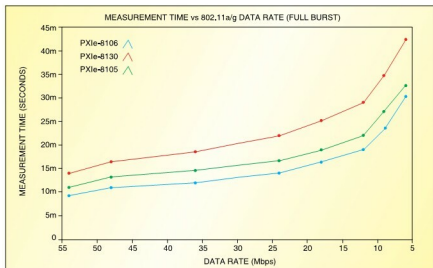


Fig. 1: Difference in measurement speed for different microprocessors (Courtesy: National Instruments)